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AGE AND MAGNETISM OF DIABASE DYKES AND TILTING OF THE PIEDMONT

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ABSTRACT

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We have conducted a paleomagnetic and radiometric study of diabase dikes in the South Carolina Piedmont where such data is lacking. Our virtual geomagnetic pole position for 20 of 24 diabase sites in the Piedmont is 83.9°E, 66.1°N and lies near the older 195 m.y. age group of Smith and Noltimier (1979). Single K-Ar apparent ages of twelve diabase sites indicate that one half of the diabase dikes studied have ages near 195 m.y. Two of the sites, whose pole positions lie well to the east of our mean value, have apparent ages in the 180 m.y. range. Thus, while two groups appear to be present in South Carolina, it is the older group that seems to predominate. We combined our pole position with other data from diabase in the southern Piedmont (Watts, 1975; Bell et al., 1979) and have calculated a mean paleopole position of 86.1°E, 66.4°N based on 50 sites in Georgia, South Carolina, and North Carolina. The southern diabase paleopole position plots slightly off the North American apparent polar wander path, but is not significantly different for the 190 m.y. average pole of Irving (1982). These results do not provide an indication of the degree of tilting of the Appalachians in post-early Mesozoic times. However, of the two possible scenarios that can be envisioned, (1) injection into an updomed area or (2) tilting of the Appalachian surface due to post-early Mesozoic differential erosion and isostatic rebound, adjustment to account for a 10° westerly tilting of the Piedmont surface about the Appalachian trend brings our southern pole position more nearly into coincidence with the North American cratonic pole for early Jurassic times.

INTRODUCTION

King (1961, 1971) recognized that basalt flows, diabase dikes, and sills intruded metamorphic terrane along the full length of the Appalachians in eastern North America during a phase of tholeiitic magmatism. May (1971) associated them with a circum-Atlantic system of diabase (in North America, South America, and Africa) related to the tensional tectonic events which produced a series of grabens and eventually led to the opening of the central Atlantic Ocean in early Mesozoic times. May showed that the tholeiitic dikes form a radial pattern within a broad elliptical

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area when the continents were reconstructed to their pre-drift configuration (Fig. 1). The parent tholeiitic magma from the upper mantle injected or passively filled deep-seated tensional fractures just prior to the separation of the continents of Pangaea. The initiation of steady spreading in the central Atlantic is dated somewhere between 180 and 195 m. y. ago (all radiometric ages in this paper have been recalculated using constants and isotopic abundances given in Steiger and Jaeger, 1978) and is thought to be coeval with the emplacement of several of the dike systems (Dalrymple et al., 1975; Smith and Noltimier, 1979; and Sutter and Smith, 1979). Sedimentation and some igneous activity within the early Mesozoic basins

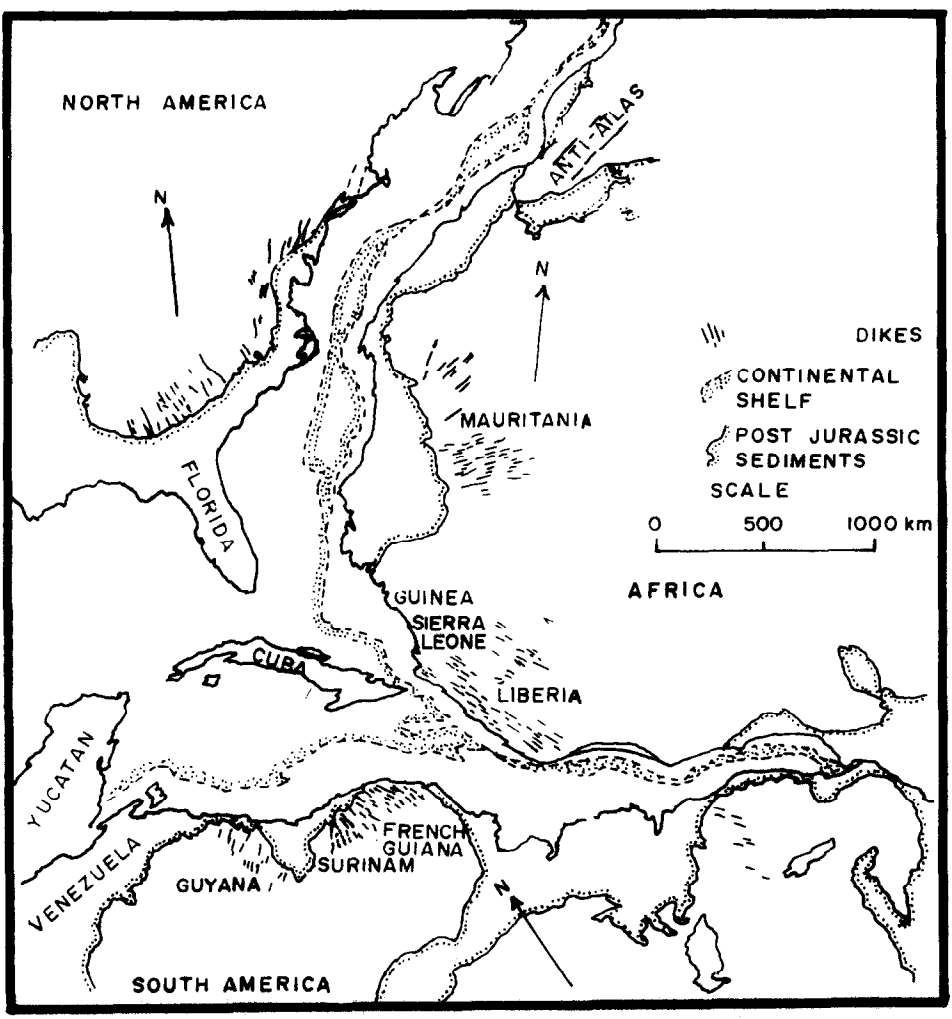


Fig. 1. Reconstruction of pre-drift of North America, South America, and Africa (after May 1971; north arrows point to present day field before reconstruction).

antedated the opening of the Atlantic and continued until sea-floor spreading at the mid-Atlantic ridge became well established. We regard these basins as early Mesozoic rather than exclusively Triassic since radiometric and faunal evidence support the presence of early Jurassic sediments and igneous rocks.

Paleomagnetic, geochronologic, and geochemical studies have been done on the diabase dikes in eastern North America and such data have been used to establish the details of tholeiitic magmatism in time and space within a tensional tectonic regime. It is convenient to divide the paleomagnetic studies into two groups: those of northeastern North America and those of southeastern North America. Most paleomagnetic work has been done for the northeastern diabase (DuBois et al., 1957; Irving and Banks, 1961; Opdyke, 1961; Larochelle and Black, 1963; Opdyke and Wensink, 1965; Larochelle and Wanless, 1966; Larochelle and Currie, 1967; Robertson, 1967; Larochelle, 1967, 1969; DeBoer, 1967, 1968; Carmichael and Palmer, 1968; Currie and Larochelle, 1969; Beck, 1972; Bowker, 1960; Barrett and Keen, 1976; Smith, 1976; Given, 1977; Volk, 1977; and Smith and Noltimier, 1979). The northeastern studies include dikes, sills and flows within early Mesozoic basins as well as dikes intruding metamorphic rocks of the Appalachians. In southeastern North America, fewer paleomagnetic studies have been carried out (Watts, 1975; DeBoer, 1968; Bell et al., 1979; DeBoer and Snider, 1979; and Phillips, in press). The study of Phillips (in press) is the only paleomagnetic study of diabase from an early Mesozoic basin in the southeast and it was from diabase recovered from three boreholes penetrating Coastal Plain sediments near Charleston, South Carolina. The rest of the paleomagnetic studies have been done on diabase dikes intruding metamorphic terrane of the southern Piedmont.

Sutter and Smith (1979) provide the most comprehensive review of radiometric studies of diabase dikes in North America and produced both K-Ar apparent ages and $^{40}Ar/^{39}Ar$ incremental release spectra of diabase in Connecticut. Many other geochronologic studies of diabase in the northern and southern Appalachians have been done (Erickson and Kulp, 1961; Fanale and Kulp, 1962; Larochelle and Wanless, 1966; Carmichael and Palmer, 1968; DeBoer, 1968; Milton and Grasty, 1969; Armstrong and Besancon, 1970; Poole et al., 1970; Baldwin and Adams, 1971; Larochelle, 1971; Corrigan, 1972; Reesman et al., 1973; Watts and Noltimier, 1974; Dallmeyer, 1975; Barnett, 1975; Deininger et al., 1975; Wampler and Dooley, 1975; Gohn et al., 1977, 1978; Houlik and Laird, 1977; Dooley and Wampler, 1977, 1978, in press; McHone, 1978; Hayatsu, 1979; Hodych and Hayatsu, 1980; Lanphere, in press).

The opening of the Atlantic Ocean in early Jurassic (Smith and Noltimier, 1979; Sutter and Smith, 1979) times may have been preceeded by a tensional updoming and crustal thinning of Pangaea similar to that proposed by Ross and Schlee (1973) for the Afar region in Africa. Such uplift would have resulted in the westward tilting of the continental margins of North America while subsequent differential erosion and isostatic rebound resulted in an eastward tilting of the Piedmont surface since

post-Mesozoic times (Zimmerman, 1979). We undertook this study of paleomagnetism and conventional K-Ar geochronology of South Carolina diabase dikes in the attempt (1) to provide paleomagnetic and geochronological data for a portion of the diabase in the southern Piedmont where such information is sparse, and (2) to see if evidence for the degree and sense of tilting of the Piedmont surface could be deduced from paleopole positions.

GEOLOGIC SETTING

Several hundred diabase dikes intrude the Piedmont province of South Carolina (Fig. 2). Many of them have not been mapped in detail and it is probable many more have yet to be mapped. A similar situation exists in Georgia and North Carolina where many early Mesozoic diabase dikes are known. Numerous geologic studies in the southern Piedmont report the occurrence of diabase dikes (Pogue, 1910; Bayley, 1928; Adams, 1938; Lester and Allen, 1950; Herrmann, 1954; King, 1961, 1971; Weigand, 1970; Bryant and Reed, 1970; Hatcher, 1971; Steele, 1971; Burt et al., 1978). Privett (1963), Overstreet and Bell (1965a,b), Butler and Siple (1966), Wagener (1970), McSween (1972), Chalcraft (1976), Bourland and Farrar (1980), Horton (1981), Peck (1981), and Smith (1982) have noted the occurrence and studied various aspects of the geology of diabase dikes in the South Carolina Piedmont. The only compilation (at 1:250,000) of diabase localities in the South Carolina Piedmont is found in the map of Overstreet and Bell (1965a). Subsequently many more diabase dikes have been discovered and mapped (for example, see Popenoe and Zietz, 1977; Daniels and Zietz, 1978; Bell et al., 1979; Peck, 1981; and Smith, 1982).

The diabase dikes in South Carolina are steeply dipping, unmetamorphosed, olivine normative tholeiites (Privett, 1963; Weigand and Ragland, 1971; Chalcraft, 1976) with an average trend in the range N40°W to N20°W, although some dikes in the northeastern part of the South Carolina Piedmont have a more nearly N–S trend. Typically, diabase is highly weathered and outcrops form trains of spheroidal boulders. The dikes intrude metamorphic host rocks of the Kiokee, Carolina Slate, Charlotte, Kings Mountain and Inner Piedmont belts of South Carolina and have been mapped cutting several 300 m.y. intrusions (Fullagar and Butler, 1976, 1978; Fullagar and Kish, 1981; and Fullagar, 1981) such as the Winnsboro, Liberty Hill, Pageland, York and Clover plutons (Wagener, 1970; Burt et al., 1978; and Bourland and Farrar, 1980). As yet none have been found cutting across the Brevard zone in South Carolina as they do in Georgia (Lester and Allan, 1950) and North Carolina (Bryant and Reed, 1970). However, several dikes cut across fault zones and boundaries between the various belts in the South Carolina Piedmont (Simpson, 1981; Secor et al., 1981), thus post-dating the last geologic processes responsible for the formation of the geologic belts in the Piedmont.

No diabase dikes have been mapped intruding the Coastal Plain sediments which rest with an angular unconformity upon the crystalline rocks of the southern

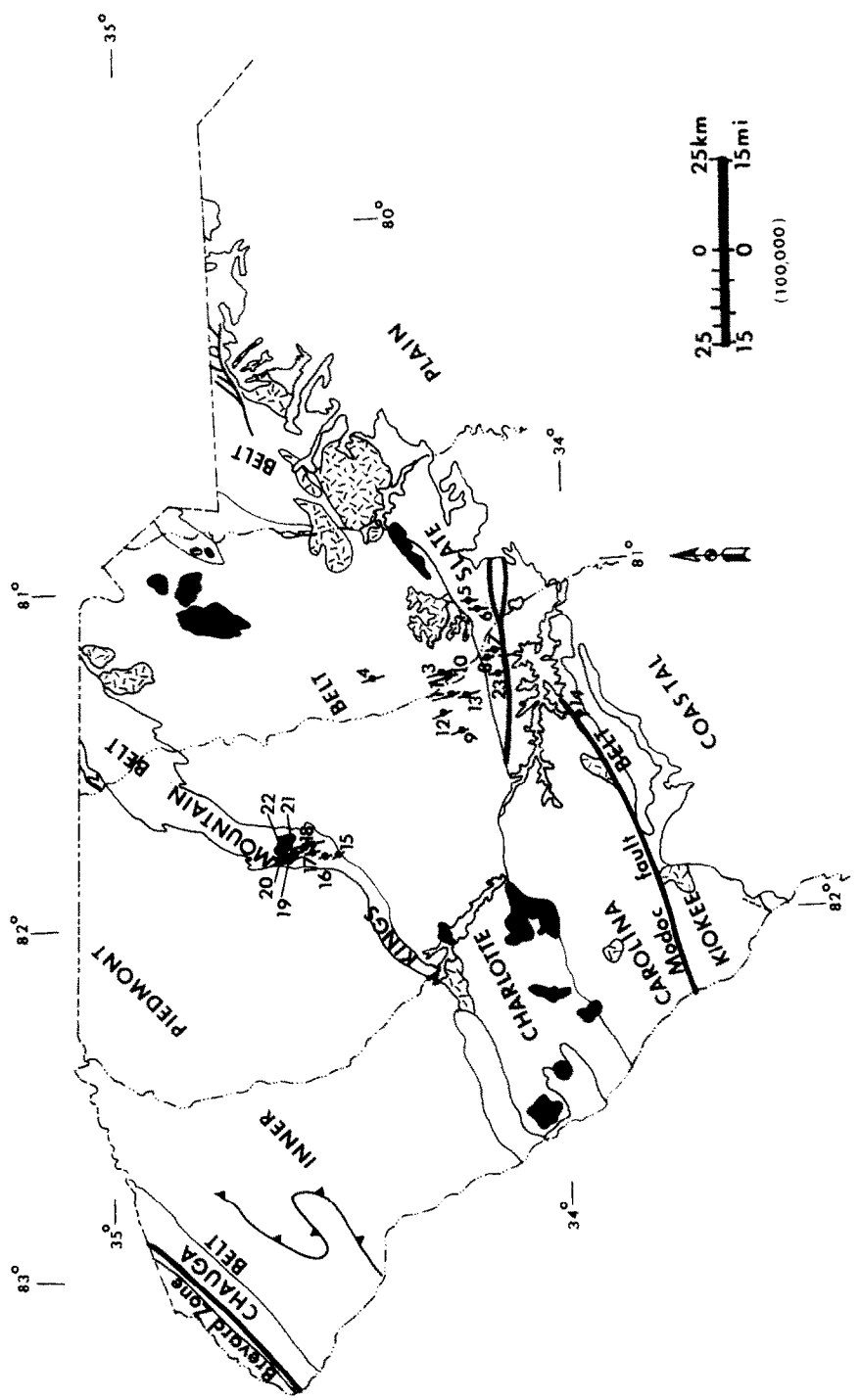


Fig. 2. Diagrammatic distribution of diabase dikes sampled in this study. (Modified after Williams, 1978.)

Piedmont. However, evidence from the U.S.G.S. boreholes near Charleston which reach diabase and basalt flows and early Mesozoic sediments at a depth of 750 m (Gohn et al., 1977, 1978), indicate their presence under the Coastal Plain.

DeBoer and Snider (1979) have summarized much of the available geochemical studies of diabase in eastern North America and correlated them with magnetic variations (Roberts, 1923; Schnabel, 1968; Ragland et al., 1968; Lapham and Saylor, 1970; Weigand, 1970; Weigand and Ragland, 1971; Steele, 1971; Rose and Smith, 1974; Smith et al., 1975; Chalcraft, 1976; Dooley, 1977; Gottfried et al., 1977, in press; Meadows, 1978). Most dikes and sills range in texture from subophitic to ophitic, and range in grain size from fine to coarse. Some of them have porphyritic to hypidiomorphic-granular textures. The diabase consists of essential plagioclase (40–60%) and augite (25–45%), and either euhedral to subhedral olivine (0–20%) or sagenetic intergrowths of quartz and K-feldspar (0–10%). Ilmeno-magnetite is the major accessory mineral present (1–3%). Biotite, sericite, chlorite, serpentine, iddingsite, and magnetite have also been identified in the diabase as secondary minerals (Weigand and Ragland, 1971; DeBoer and Snider, 1979).

Weigand and Ragland (1971) divided the tholeiitic igneous rocks into four geochemical groups: olivine normative tholeiite, low and high TiO_2 quartz normative tholeiites, and high Fe_2O_3-high TiO_2 quartz normative tholeiite. They have noted a geographic distribution for the volume of the geochemical groups, with quartz normative tholeiites predominating the northeastern Appalachians and olivine normative tholeiites most abundant in the southeast. The quartz normative composition can be derived from olivine normative parental magma by removing various mineral assemblages through fractional crystallization, but not the converse, suggesting that the quartz tholeiites are the younger magmas (Weigand and Ragland, 1971). Smith et al. (1975) found field evidence that suggests olivine tholeiites intruded early in the tensional tectonic regime, followed by emplacement of high TiO_2 quartz magmas. Weigand and Ragland (1971) have used the hypothesis that crustal contamination played a major role in determining chemical composition of the magma during ponding of parental magma in intermediate levels of the crust, while olivine normative magmas ascended more rapidly to the site of intrusion with little or no crustal contamination.

In South Carolina, quartz normative tholeiites have been found in buried early Mesozoic basins (Gottfried et al., 1977), while most of the diabase dikes that intrude the South Carolina Piedmont are olivine normative tholeiites (Chalcraft, 1976). DeBoer and Snider (1979) have used the more Mg-Fe rich nature of the olivine tholeiites (found in the Piedmont) in South and North Carolina to suggest that the Carolinas were the site of an early Mesozoic hot spot.

ANALYTICAL TECHNIQUES

Twenty-four diabase sites were drilled with a portable, water-cooled, field drill fitted with diamond bits to obtain between six and eight 25 mm (± 0.5) diameter

cores from each of the diabase sites within the South Carolina Piedmont. Their distribution is diagrammatically shown in Fig. 2. Detailed magnetometer surveys to map the occurrence and extent of some of these dikes is in progress (D.T., Secor et al., pers. commun., 1981). Sampling was spread as far apart as each outcrop allowed and most cores were separated by one meter or more. In a few instances cores were drilled as close together as 0.5 m. The orientation of each core was measured with an aluminum orienting device and an azimuthal compass. The core specimens were sliced into one or more 22.2 mm (± 0.5) length cylinders with a diamond saw. The intensity, declination, and inclination of each specimen were measured with a Digico complete results magnetometer interfaced with a Digital MINC 11 minicomputer.

Following NRM direction measurements, one pilot specimen from each site was progressively demagnetized in alternating fields in 5 mT increments from 5 to 50 mT, and also at 75 mT and 100 mT values. Also one pilot from each site was progressively demagnetized thermally in 25°C increments from 100° to 200°C and in 50°C increments from 250° to 650° increments. Stable end point curves and demagnetization spectra were used to determine the stage at which the characteristic magnetization was isolated. This usually occurred after demagnetization at 15 mT, but in a few cases the characteristic magnetization field only appeared after demagnetization in fields as high as 30 to 35 mT.

A representative slice (5 mm thick) of a sample core from each of twelve sites was used for conventional K-Ar age determinations. Details of the mass spectrometric analysis and argon extraction line at Georgia Tech are described in Dooley and Wampler (in press).

Each sample was crushed and pulverized to 2 mm sized aggregates, retaining the fine powder, and was separated by a sample splitter into two aliquots for potassium and argon analyses. The potassium aliquot was dissolved in a perchloric and hydrofluoric acid solution and analysed along with liquid standards (1–5 ppm potassium) on an atomic absorption spectrophotometer. The argon aliquot was weighed into copper capsules, vacuum dried, and loaded into an argon extraction system, on-line with a MS 10 gas mass spectrometer. The sample was then melted and evolving gases purified before being admitted into the mass spectrometer in the static mode (Dooley and Wampler, in press).

DISCUSSION

Demagnetization spectra

Typical alternating field (AF) and thermal demagnetization spectra are shown in Fig. 3. The NRM intensities were in the range $8.0 \cdot 10^{-2}$ to $9.9 \cdot 10^{-5}$ A/m, averaging about $7.1 \cdot 10^{-3}$ A/m. The AF curves of Fig. 3 show that in the majority of cases, 40–60% of the NRM intensity vanishes at 5 mT, and that the characteristic

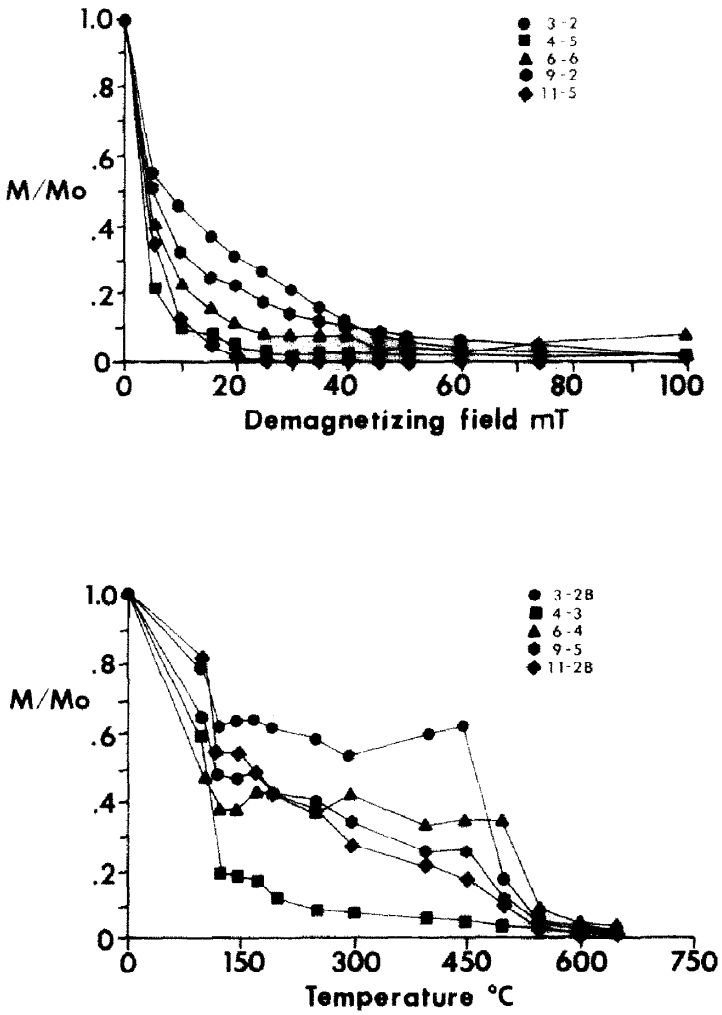


Fig. 3. Alternating field (mT) and thermal (°C) demagnetization spectra for representative samples.

magnetization appears when 80–90% of the NRM has been removed between 15 and 25 mT.

Such magnetic behavior (appearance of characteristic magnetization by 15–25 mT, low coercivity) is typical of tholeiitic Newark trend basalts of the eastern North American Piedmont (DeBoer, 1968; Beck, 1972; Watts, 1975; Smith, 1976; Smith and Noltimier, 1979; and DeBoer and Snider, 1979). Diabases commonly have characteristic magnetic moments carried by a mixture of multidomain and single domain minerals (Johnson and Merill, 1973; Stacey and Bannerjee, 1974).

It is interesting that whereas in AF demagnetization the stable direction is

revealed at 15 mT (for most sites), it appears necessary to demagnetize at temperatures in the range 350°–450°C to recover the characteristic magnetization by thermal demagnetization (Fig. 3). The Curie point of these rocks lies somewhere within the range 500°–550°C and there is evidence of a blocking temperature of about 150°C in some but not all of the dikes.

Stable end point curves (similar to Zijderveld diagrams) have been plotted for three sites (3, 4, and 9). Figure 4 displays a stable end point curve from a diabase dike (site 9) similar to other Appalachian dike results in which magnetite is the major magnetic carrier (Beck, 1972; Johnson and Merill, 1973; Stacey and Bannerjee, 1974; Smith, 1976; Smith and Noltimier, 1979). Both AF and thermal stable end point curves decrease linearly to zero for both declination and inclination which is typical behavior during progressive demagnetization of basaltic rocks.

Sites 3 and 4 (Figs. 5 and 6) exhibit similar demagnetization behaviors during AF and thermal demagnetization. However the thermal demagnetization for these two sites was different than that for site 9 (Fig. 4). The AF stable end point curves for sites 3 and 4 show that a viscous magnetic component is removed between 5 and 15 mT. The stable end point curves then approach zero linearly over the demagnetizing field range of 20–75 mT. The thermal stable end point curves for both sites 3 and 4 show that between 150° and 450°C the directions of magnetic remanence and intensity vary little. At 450°C the directions of magnetization and intensity begin to decrease in a more erratic fashion than during AF demagnetization.

Directions

The site mean directions of magnetization and the standard statistical parameters are listed in Table 1. When a single core had directions of magnetization well removed from that of the rest of the cores of a particular site this sample was excluded from the computation of site mean directions, for example one sample was ommitted from site 8 (Table 1). The cause may arise from sampling errors or reflect some chemical weathering effect.

The effect of magnetic cleaning is contrasted in Fig. 7 with the NRM and cleaned directions plotted for three sites. Only the cleaned directions are listed in Table 1. In contrast to the tight NRM clusters for dikes in the northeastern Appalachians (Smith and Noltimier, 1979) the NRM directions of our samples are highly dispersed both in declination and inclination. A secondary viscous magnetic component (probably due to the present magnetic field of the earth) may be responsible for such dispersion and although our NRM intensities are relatively low compared to those typically induced by lightning, the effect of lightning remagnetization cannot be ruled out.

The cleaned directions of magnetization show a distinct clustering although the alpha 95 values tend to be higher than those reported from similar dikes collected from the northern Appalachians (DeBoer, 1968; Beck, 1972; Smith, 1976; Smith and

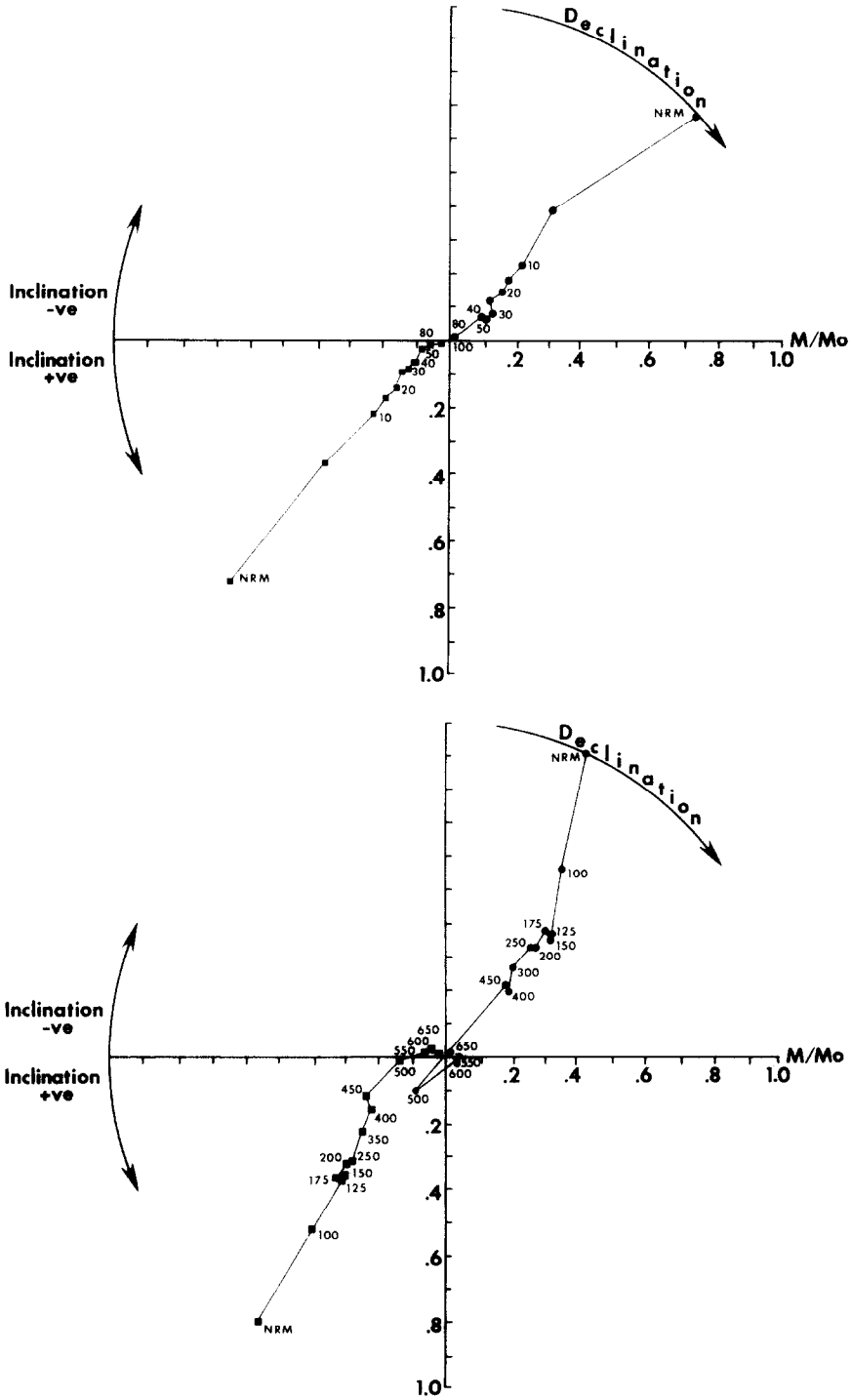


Fig. 4. Stable end point curves (site 9) typical of Newark Trend basalts for thermal (bottom, °C) and AF demagnetization (top, mT).

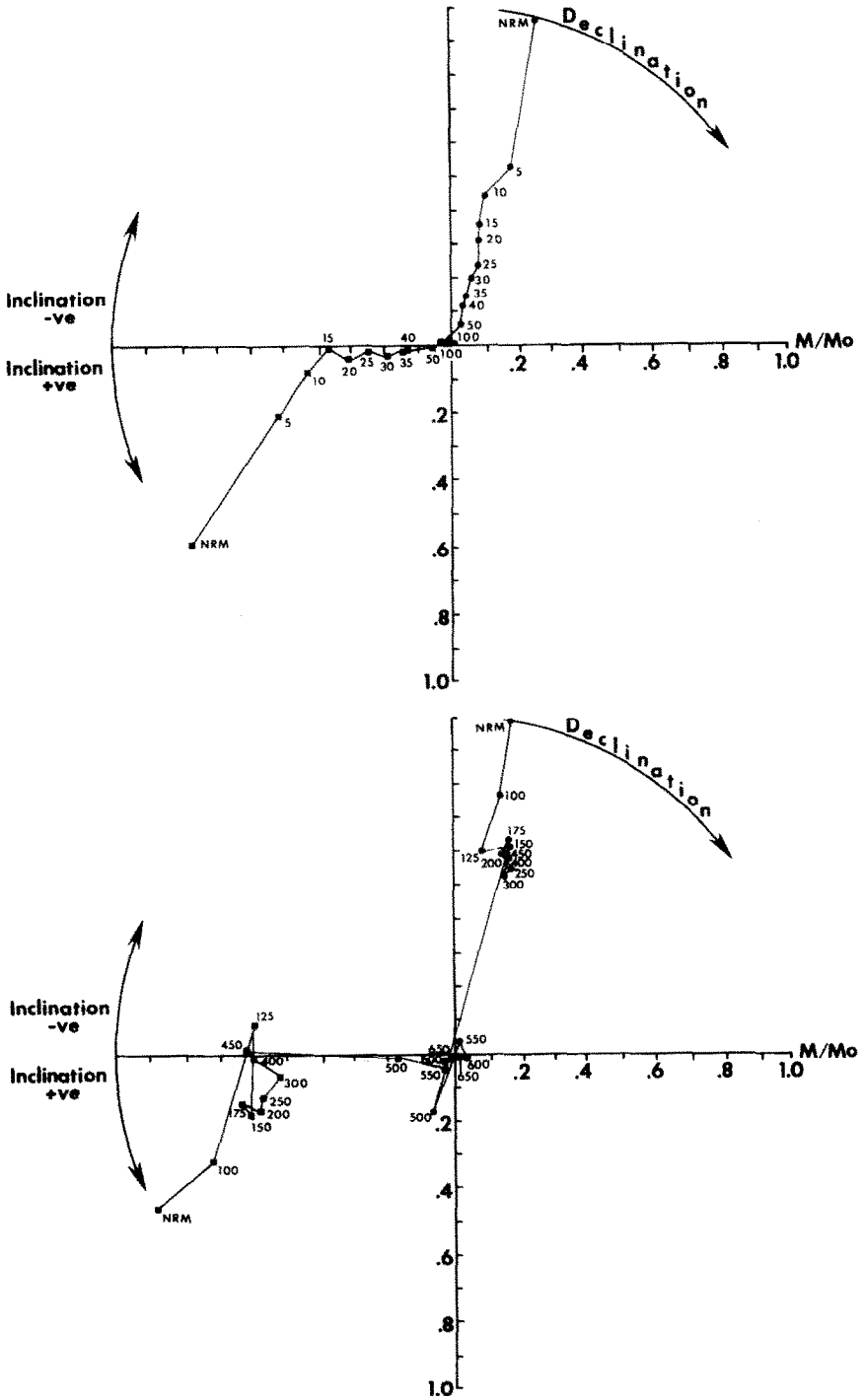


Fig. 5. Stable end point curves for thermal (bottom, °C) and AF demagnetization (top, mT) of site 3.

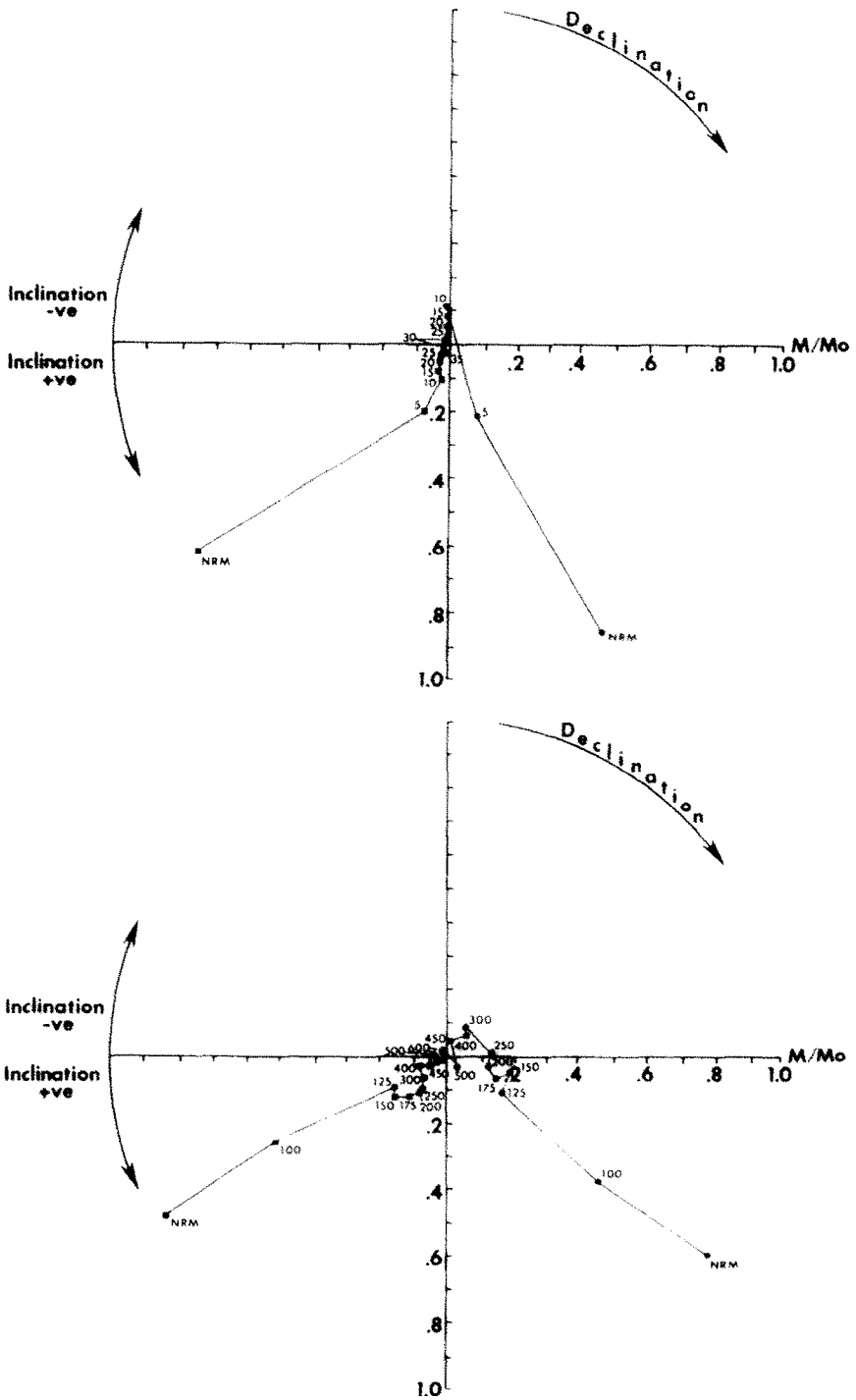


Fig. 6. Stable end point curves for thermal (bottom, °C) and AF demagnetization (top, mT) of site 4.

TABLE 1 SUMMARY OF STATISTICAL PARAMETERS

SITE	DEC°	INC°	SAMPLING a/b	α95	K	R	DEMAGNETIZATION	VGP LAT°N	VGP LONG°E	[3] K-Ar apparent age	LOCATION LAT.	LOCATION LONG.
[+]3	9.5	0.5	7/7	6.0	101.5	6.9409	AC 15 mT	54.8	97.9	205 m.y.	34.3	-81.3
4[1]S	8.9	34.6	6/6	13.3	25.1	5.8004	AC 25 mT	72.6	110.6	288 m.y.	34.5	-81.3
[2]B	12.2	36.8	4	17.3	29.2	3.8971	AC 25 mT					
5 S	6.3	27.6	5/8	19.0	17.3	4.7682	AC 30 mT	69.6	98.9	202 m.y.	34.2	-81.1
B	7.0	29.2	4	*25.9	13.5	3.7781	AC 30 mT					
6 S	19.3	-54.1	6/6	*32.8	5.1	5.0235	AC 25 mT, 30 mT			194 m.y.	34.2	-81.1
B	9.9	-60.8	3	*47.1	7.9	2.7474	AC 25 mT, 30 mT					
7 S	16.8	-12.4	6/7	*24.0	8.8	5.4298	AC 30 mT, 450°C.			191 m.y.	34.2	-81.3
B	19.8	-8.1	5	*27.2	8.0	4.5478	AC 30 mT, 450°C.					
8 S	12.1	12.4	5/6	12.3	39.5	4.8986	AC 15 mT	59.9	105.9	193 m.y.	34.2	-81.3
B	14.3	19.8	4	14.0	43.9	3.9317	AC 15 mT					
9 S	40.8	35.8	6/6	8.0	71.6	5.9301	AC 20 mT, 35 mT	51.2	160.5	180 m.y.	34.2	-81.5
B	40.0	36.5	5	8.6	79.3	4.9436	AC 20 mT, 35 mT					
10 S	9.7	16.3	8/8	12.4	30.3	7.5196	AC 35 mT, 450°C., 500°C.	62.5	102.5	194 m.y.	34.3	-81.3
B	15.7	18.1	4	17.6	28.3	3.8939	AC 35 mT, 450°C., 500°C.					
11 S	340.9	6.0	5/7	15.1	26.6	4.8495	AC 30 mT, 500°C.	54.0	47.5	195 m.y.	34.3	-81.3
B	342.3	18.8	3	15.3	66.0	2.9637	AC 30 mT, 500°C.					
[+]12 S	350.4	35.7	7/7	8.3	54.0	6.8889	AC 15 mT	73.2	48.6	194 m.y.	34.3	-81.4
13 S	14.0	28.7	8/8	16.6	12.1	7.4231	AC 15 mT	67.2	118.3	207 m.y.	34.3	-81.2
B	19.6	27.5	5	16.8	21.8	4.8166	AC 15 mT					
14 S	32.9	58.0	5/6	*21.2	14.0		AC 30 mT			184 m.y.	34.0	-81.5
B	26.9	48.8	2	----	----	1.8437	AC 30 mT					
15 S	358.1	25.7	5/7	8.4	84.8	4.9529	AC 20 mT	68.8	77.0		34.6	-81.8
B	357.3	26.3	4	5.7	262.9	3.9886	AC 20 mT					

TABLE 1 CONTINUED

SITE		DEC°	INC°	SAMPLING a/b	α95	K	R	DEMAGNETIZATION	LAT°N	LONG°E	NOTES	LAT.	LONG.
17	S	31.4	26.9	5/7	10.6	52.7	4.9241	AC 20 mT	55.1	154.0		34.7	-81.8
	B	31.1	24.6	3	9.4	9.4	2.9886	AC 20 mT					
18	S	0.2	32.9	7/10	7.1	73.8	6.9187	AC 10 mT	73.3	82.6		34.6	-81.8
	B	358.6	33.3	5	6.7	130.3	4.9693	AC 10 mT					
19	S	4.8	-6.9	5/6	*91.6	-----	2.6021	AC 25 mT				34.7	-81.9
	B	19.7	1.0	2	*44.9	-----	1.0300	AC 25 mT					
20	S	341.7	27.3	8/8	16.9	11.7	7.4026	AC 25 mT	63.9	38.1		34.7	-81.9
	B	344.4	24.7	6	19.6	12.6	5.6038	AC 25 mT					
21	S	358.4	31.5	7/8	8.2	54.8	6.8904	AC 25 mT				34.7	-81.8
	B	357.9	30.3	3	13.9	78.2	2.9747	AC 25 mT					
†23	S	352.5	21.4	9/9	7.7	45.6		AC 15 mT	65.9	65.8		34.2	-81.3
24	S	3.9	23.6	---	2.9	144.0	16.8890	AC 30 mT	67.2	88.4	[4]Previous work		
25	S	351.0	8.2	---	3.7	----	---	AC 25 mT	58.9	116.4	Previous work		
26	S	356.3	-14.2	---	6.7	98.0	5.9490	AC 25 mT	63.0	107.0	Previous work		
27	S	350.3	-17.1	---	7.1	53.0	8.8490	AC 20 mT	63.1	120.5	Previous work		

MEAN POLE DIRECTIONS

	DEC°	INC°	LAT°N	LONG°E	SAMPLING a/b	α95	K	R	dp	dm	Comments
S	1.0	20.0	66.1	83.9	20/24	8.9	14.5	18.6851	4.9°	9.3°	This value is taken as our best average for a virtual geomagnetic pole position. Average VGP for sites giving unit weight to blocks (two or more core/block). Same 12 sites in "B" calculation for comparable results.
B	4.9	27.7	69.7	95.3	12/24	10.6	17.7	11.3792			
B	3.5	26.0	69.0	91.1	12/24	11.1	16.4	11.3287			

[1]S Mean declination and inclination giving unit value to individual core samples

[2]S Mean declination and inclination giving unit value to several cores in individual blocks

[3]K-Ar apparent age determinations based on preliminary single analyses of each site; Decay constants $\lambda_e = 0.581 \times 10^{-10} y^{-1}$, $\lambda_\beta = 4.962 \times 10^{-10} y^{-1}$, $^{40}K/K_{total} = 0.000167$, (Steiger and Jaeger, 1977 K-Ar Conventional analyses done at the Georgia Institute of Technology, School of Geophysical Sciences

[4]Previous paleomagnetic work done at the University of South Carolina under the direction of Dave Martin and Alan Nairn

Sampling: a = number of samples used in calculation; b=number of samples measured

* α95 values greater than 20° have not been used in pole position calculations

α95 = radius of cone of 95% confidence

†These sites have an S column only because no two cores were taken from an individual block

K = Fisherian precision parameter (Fisher, 1953)

R = Vector sum of unit magnetizations of specimens used in calculations

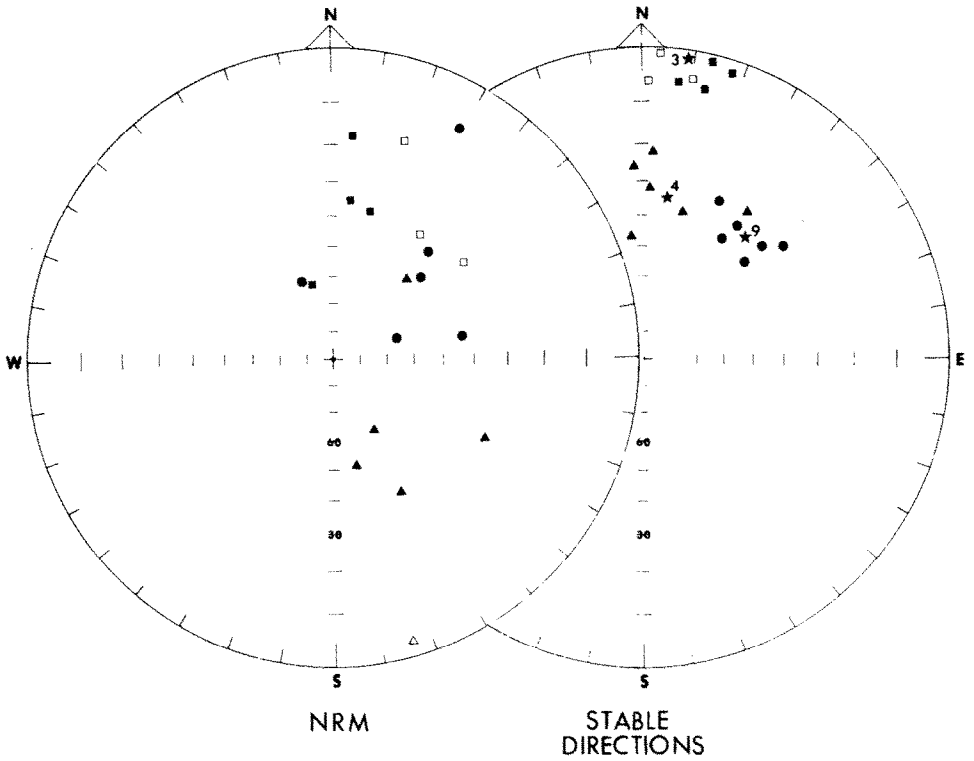


Fig. 7. NRM directions contrasted with cleaned directions for three sites.

Noltimier, 1979), but similar to those reported by Bell et al. (1979) from localities in the South Carolina Piedmont. The mean values for all dikes are listed in Table 1, but in the computation of a mean direction of magnetization we have chosen to exclude those site mean directions of magnetization with alpha 95 values in excess of 20° rather than the 10° value used by Smith and Noltimier (1979). This resulted in the rejection of four sites (6, 7, 14 and 19, in Table 1).

Although the basic dikes in the Southern Appalachians undergo severe weathering and many outcrops may only appear as a train of spheroidal boulders, this does not appear to result in any pronounced rotation of individual blocks (for most sampling sites). We have tested this idea by calculating means for sites giving unit weight to cores (S column in Table 1) and to blocks (B column in Table 1) combining the data where there are more than one core per block. As the data in Table 1 indicate, the dispersion seems to remain unchanged between the two different calculations for the majority of sites. Bell and others (1979) report a similar observation.

POLE POSITION

In Table 1, our virtual geomagnetic pole position (VGP) based on 20 of 24 sites (incorporating earlier results) is 83.9°E, 66.1°N (Fig. 8). It lies within the limits of other early Jurassic pole positions for diabase in the Appalachians (Fig. 9) as well as those from stable cratonic North America (Table 2). There is no clear indication of two groupings such as the two pole positions of Smith and Noltimier (1979; 83.2°E, 63.0°N for the 195 m.y. age group; and 103.2°E, 65.3°N for the 180 m.y. age group). Our VGP lies close to the older 195 m.y. group. This observation has been tested by preliminary conventional K-Ar apparent age determinations. Seven of twelve dikes (sites 6, 7, 8, 9, 10, 11 and 12) have an average apparent age (based on single analysis of each site) of 194 ± 4 m.y. and all of them are near this average (Table 1). Three diabase sites have single apparent ages near 205 m.y., two sites near 182 m.y., and one had a single apparent age of 288 m.y. There are two sites (9 and 14) in which the declination is considerably to the east of that of the remainder and by coincidence these are the two sites yielding apparent ages in the 180 m.y. range.

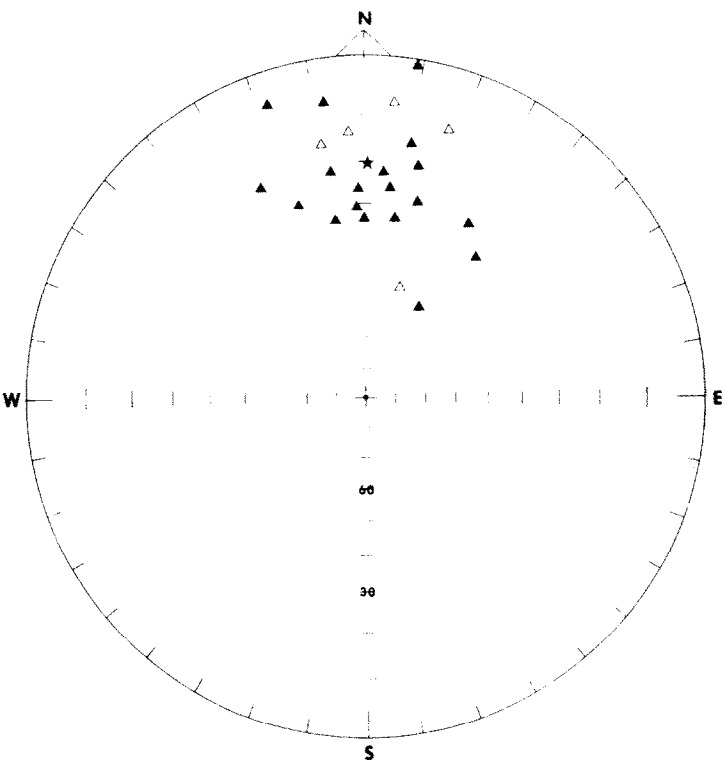


Fig. 8. Mean declinations and inclinations of 24 diabase sites in the South Carolina Piedmont (open triangles are negative inclinations).

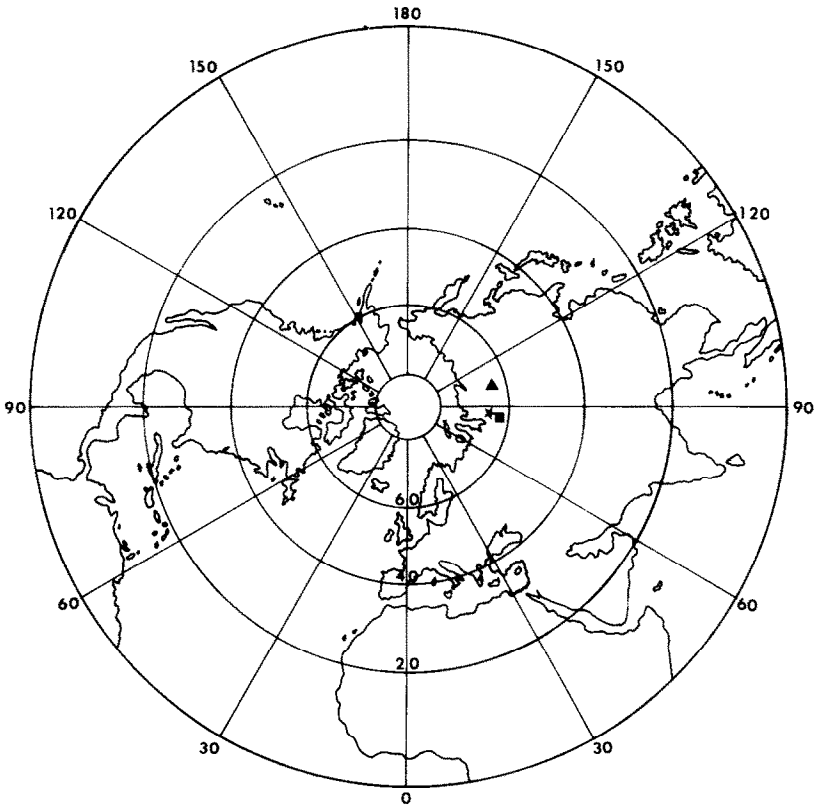


Fig. 9. Pole positions of early Mesozoic diabase from eastern North America (the star represents the southern diabase pole, the triangle represents the pole position of the 195 m.y. age group and the square represents pole position for the 180 m.y. age group of the northern diabase, Smith and Noltimier, 1979).

There is indication that many of the eastern North American diabases have been contaminated by at least small amounts of excess ^{40}Ar (Sutter and Smith, 1979) and in particular, large amounts (comparable to the results of diabase from Liberia; Dalrymple et al., 1975) have been found in diabase dikes that intrude the Piedmont of Georgia (Dooley and Wampler, 1977; in press). The anomalous 288 m.y. apparent age for one dike in this study is probably a result of excess ^{40}Ar contamination, while $^{40}Ar/^{39}Ar$ incremental release spectra will be required to be certain that the rest of the K-Ar apparent ages in this study do not reflect the presence of excess ^{40}Ar in small amounts. Thus, it seems that while the older diabase dikes (195 m.y. group) are relatively common in the South Carolina Piedmont, the younger group (180 m.y.) appears to be much less common on the basis of sparse paleomagnetic and geochronologic data.

Our virtual geomagnetic pole position (83.9°E, 66.1°N) can be combined with other VGP's from the southern Piedmont (Watts, 1975; Bell et al., 1979) to yield a

TABLE 2

North American Late Triassic (?)-Early Jurassic Paleomagnetic Data

Unit	Age	N	K	α95	dm	dp	ELong.	N.Lat.	Source*
Kayenta fm. Utah	$T_RU(?)$	105	32	2.5	2.6	1.4	72.7	61.7	Steiner and Helsey (1974)
Kayenta fm. Utah	$T_RU(?)$	7	102	6.0	-	-	82.5	61.2	Johnson (1976)
Group I NE North America		72	56	2.3	-	-	83.2	63.0	Smith and Noltimer (1979)
Group 2 NE North America		156	92	1.4	-	-	103.2	65.3	Smith and Noltimier (1979)
Summerville fm. Utah	Jm Callovian	15	71	4.6	5.4	3.2	110.6	67.5	Steiner (1978)
Topley fm.	Jm	13	12	9.1	14.4	11.4	128.6	72.0	Symon (1973)
Morrison fm. Utah (Lower)	Ju Kimmeridgian	32	23	5.4	6.7	4.7	143.0	61.4	Steiner and Helsey (1975)
Morrison fm. Utah (Upper)	Ju	68	35	3.0	4.2	3.0	163.8	67.8	Steiner and Helsey (1975)

* Original references found in Smith and Nottimier (1979).

mean paleomagnetic pole position of 86.1°E, 66.4°N (A95 = 3.12°, $K = 42.79$, $R = 48.8550$, Table 3). This paleoposition is slightly to the west of the North American polar wander path but is not significantly different from the 190 m.y. average of Irving (1982) (Fig. 9). The angular distance between the 190 m.y. pole of Irving (1982) and the southern diabase paleopole is no more than 6°–10°. These results cannot provide any indication of the degree of tilting of the Appalachians, since they are not statistically different. Yet of the two possible cases that can be envisioned, (1) injection into an updomed area or (2) tilting of the surface due to post-Mesozoic differential erosion and isostatic rebound, adjustment to account for the former case (1) brings our southern pole position into coincidence with the North American cratonic pole.

Tilting of the Appalachian surface since early Mesozoic times must have been complex. In the initial stages of continental rifting, updoming was the dominant motion of the surface and tilting to the west about the Appalachian trend would have resulted. Subsequently the Appalachian surface has undergone rapid erosion in some areas (for example, an average rate of erosion of 0.031 mm/yr since early Jurassic times for the northern New England Piedmont has been determined by Doherty and Lyons, 1980). The topographically high Appalachian surface is evidence for continuing isostatic readjustment occurring in the are today (Roper, 1980). This has led to differential erosion of the Piedmont surface, the greatest amount

Table 3

Mean Pole Positions of Eastern North American early Mesozoic Igneous Rocks

Northeastern United States and Canada

N	K	α_{95}	E. Long^{o}	N.Lat^{o}	Reference
52	104	1.9	98.9	68.4	Smith and Noltimier, 1979
25	107	2.8	82.5	63.1	Smith and Noltimier, 1979
59	68	2.3	83.9	62.6	Smith and Noltimier, 1979

Southeastern United States

N	K	α_{95}	E. Long^{o}	N. Lat^{o}	Reference
25	-	7.7	80.1	64.9	Watts, 1975 [1]
5	22.0	6.3	83.0	62.0	Bell and others, 1980[2]
20	14.5	8.9	83.9	66.1	This study[3]

N	K	α_{95}	E. Long^{o}	N.Lat^{o}	
Mean Pole Position for Northeastern United States					
72	56	2.3	83.2	63.0	Smith and Noltimier, 1979[4]
156	92	1.4	103.2	65.3	Smith and Noltimier, 1979[5]
Mean Pole Position for Southeastern United States					
50	48.9	3.1*	86.1	66.4	

*This value was determined using VGP's of each site rather than Declination and Inclination and is an A95 rather than an α_{95} and is a circle of confidence.

[1]Pole position taken from Watts ms using data from Georgia and North Carolina.

[2]Pole position taken from Bell and others (1980) from dikes in the Haile-Brewer Area of South Carolina.

[3]Pole position taken from this study based on 20 sites collected in the South Carolina Piedmont.

[4]Based on a calculation by Smith and Noltimier (1979) using previously published pole position: This pole corresponds to Group I of Smith and Noltimier's which was dated as a 195 m.y. group by Sutter and Smith, 1979.

[5]Based on a calculation by Smith and Noltimier (1979) using previously published pole positions: This pole position corresponds to Group II of Smith and Noltimier's and was dated as 180 m.y. group.

occurring along the western edge of the Piedmont and the least along the eastern edge (Zimmerman, 1979). Such differential erosion rates can be used to infer an eastward tilting of the Appalachian surface since post-Mesozoic times. Thus, the resultant between the westward (case 1) and eastward (case 2) tilting of the Appalachians would depend upon which motion was greatest.

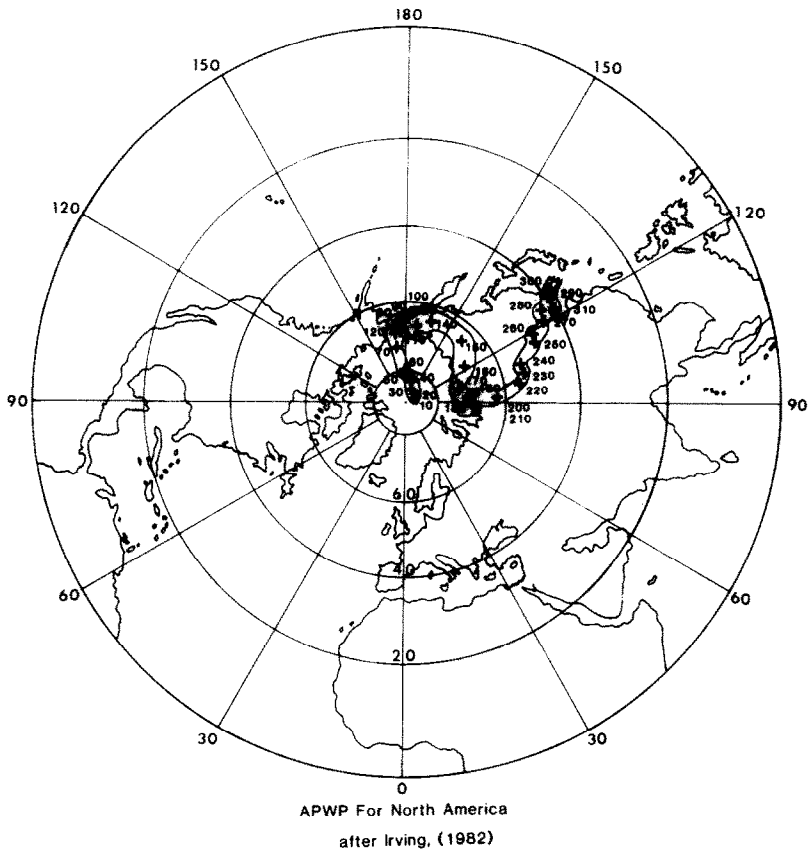


Fig. 10. Southern paleopole position calculated in this study of early Mesozoic diabase plotted relative to the Paleozoic North America apparent polar wander path (mean poles for 10 m.y. intervals are marked by crosses, Irving, 1982).

resultant between the westward (case 1) and eastward (case 2) tilting of the Appalachians would depend upon which motion was greatest.

This idea can be tested by assuming a small degree of tilting about the Appalachian trend that would be necessary to bring the southern pole position into agreement with the 190 m.y. pole of Irving (1982) (Fig. 10). Applying a 10° westerly tilt correction to the Appalachian surface brings the pole position almost into coincidence with the 190 m.y. position of Irving (1982). Applying the same correction to the east results in a pole position significantly different from the North American polar wander path for early Mesozoic times. This suggests that the differential erosional tilt effect (case 2) is outweighed by the presumed domal effect (case 1). While our pole position cannot prove such a conclusion, we feel that the results of this study constrain the sense of tilting of the Appalachian surface to the west and that the degree of tilting due to updoming was greater than subsequent tilting due to differential erosion and isostatic adjustment.

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